

Cisco Packet Tracer Lab Report

Lab Name: Lab 01 – Build a Small Office Network

Date Completed: 19/07/2026

Category: Networking / LAN Configuration

1. Objective (2–3 sentences)

The aim of this project was to build a Local Area Network (LAN) similar to one found in a typical office environment. The learning outcome of this lab was to understand the components required to connect multiple devices within a small network, as well as the basic configuration needed to enable network communication.

2. Network Design

A small office network was created including a router, switch, and 3 PCs interconnected using Copper Straight-Through cables. The devices were configured to allow communication within the local network.

- Devices:
 - Router
 - Switch
 - 3 PCs
- Topology:
 - PCs connected to switch
 - Switch connected to router
- Cable used:
 - Copper-straight through

3. Configuration

I manually assigned an IP address to the router and 3 PCs, applying the principle of static IP addressing. This step was important to ensure that each device could be identified on the network and transfer data to one another. Also, I configured the PCs

subnet mask to 255.255.255.0 and IP addresses with the same network portion of 192.168.1.x, helping them decide how to send data, either directly or through the default gateway (the router).

Then I enabled the router's Gigabit Ethernet 0/0 port via the Command Line Interface using the `no shutdown` command, allowing the router to use the physical link established by the copper cable.

4. Verification & Troubleshooting

I tested the router interface GigabitEthernet 0/0 to see if it had been enabled by running the `show ip interface brief` command. A couple of times the console displayed that its status and protocol were down. I troubleshooted the issue by checking the cable ports on the router and switch and correcting the cable connection. After running the `show ip interface brief` command again, the router showed the status and protocol were both up. I verified that the connection was made by pinging the router from one of the PC's command prompt terminals, demonstrating that the PC could communicate with the router.

5. Key Concepts Learned

- **IP addressing** — understanding devices need IP addresses to identify each other on the network.
- **Switches and routers** — recognizing their differing roles: the switch to connect multiple devices within a network, and the router to connect different networks.
- **Static IP addressing** — one of the methods to assign IP addresses is by manually assigning an IP address and subnet mask to the PCs instead of having them automatically assigned by a DHCP server.
- **Network interfaces** — `no shutdown` is a key command to enable a router interface after configuration, with feedback of up/up and down/down showing the status and protocol response of the router interface.
- **Troubleshooting** — the importance of using the right cable and connecting to appropriate ports to establish a connection between devices.

6. Screenshots

Figure 1 – Network Topology

The completed network showing the router, switch, 3 PCs and their physical connections.

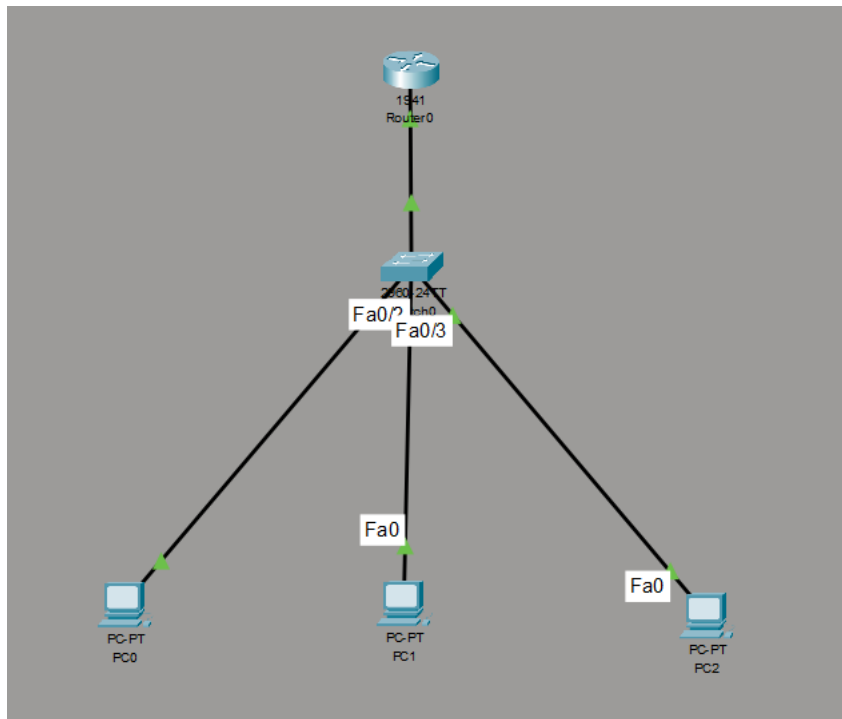


Figure 2 – Router Interface Verification

The show ip interface brief output confirming the GigabitEthernet0/0 interface is up/up.

Press RETURN to get started!

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#exit
Router#
$SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.1.1    YES manual up          up
GigabitEthernet0/1  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
Router#
```

Figure 3 – Connectivity Tests

Successful pings between a PC and the router, and between two PCs, confirming connectivity across the network.

```
C:\> ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=53ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=13ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 53ms, Average = 16ms

C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

7. Reflection (2–3 sentences)

This lab helped me understand how different network components, such as switches, routers, IP addresses, and subnet masks, work together to allow devices to communicate within a small network. I also gained practical experience with configuring and troubleshooting a network, including identifying why the router interface showed down/down and resolving the issue, which has given me a stronger foundation to build on in future networking practicals.